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A PRACTICAL- BASED LEARNING REPORT

ON

"ONLINE BUS TICKET BOOKING SYSTEM"

Submitted in partial fulfillment of the requirements for

SOFTWARE ENGINEERING [BCS755C]

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CERTIFICATE

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ABSTRACT

An online bus ticket booking system is a web-based platform designed to simplify and streamline the process of reserving bus seats through digital channels. It allows users to search for available routes, compare schedules and fares, select preferred seats, and complete secure payments without the need to visit physical booking counters. The system enhances convenience, reduces manual errors, and improves operational efficiency for both passengers and bus operators by providing real-time seat availability, automated notifications, and easy cancellation or rescheduling options. Additionally, it supports 24/7 accessibility, enabling users to book tickets anytime from any location, while offering a centralized database that helps operators manage fleets, schedules, and customer information more effectively. Overall, it delivers a fast, reliable, and user-friendly alternative to traditional ticketing methods.

TABLE OF CONTENTS

SL. NO	NAME	PAGE NO
1	INTRODUCTION	01-03
2	LITERATURE SURVEY	04
3	SYSTEM SPECIFICATION	05-07
4	DESIGN	08-12
5	IMPLEMENTATION	13-15
6	RESULT	16-18
	CONCLUSION	19
	REFERENCES	20

FIGURE OF CONTENT

FIG NO	NAME	PAGE NO
4.1	USE CASE DIAGRAM	08
4.2	ER DIAGRAM	09
4.3	CLASS DIAGRAM	10
4.4	SEQUENCE DIAGRAM	11
6.1	SNAPSHOTS	17

CHAPTER – 1

INTRODUCTION

Transportation plays a crucial role in the overall growth and development of any country, and among various modes of transport, bus travel is one of the most affordable and widely used means, especially in developing nations like India. With the rapid expansion of digital services, people increasingly expect faster, easier, and more efficient ways to access transportation facilities. Traditionally, booking a bus ticket required passengers to physically visit a bus depot, stand in long queues, or depend on travel agents. This manual system was time-consuming, prone to errors, and lacked transparency. In recent years, the emergence of online booking systems has transformed the travel industry by making the entire process more convenient and user-friendly. An Online Bus Ticket Booking System is one such technological advancement that helps eliminate the drawbacks of the traditional method by providing a centralized, automated, and real-time platform for booking bus tickets.

This mini-project focuses on designing and developing an online bus ticket booking system using the principles of Software Engineering, emphasizing systematic development, structured analysis, and efficient implementation. Software engineering provides a disciplined approach to handle complex systems by applying engineering principles, tools, and methodologies throughout the software development life cycle (SDLC). By applying these structured methodologies, the project ensures that the system is reliable, user-friendly, scalable, and maintainable for future improvements.

The primary purpose of the Online Bus Ticket Booking System is to simplify the ticketing process for both passengers and bus operators. The system allows users to search for available buses, select suitable routes, check seat availability, make bookings, and receive confirmation instantly. This eliminates unnecessary delays and reduces dependency on physical ticket counters. In addition, bus operators can manage schedules, maintain passenger records, update bus details, and monitor bookings through a well-structured backend interface. This digitization not only enhances the efficiency of operations but also improves customer satisfaction by offering transparency, accuracy, and convenience.

The need for automation in the bus reservation system arises due to several limitations in the conventional approach. Manual ticketing leads to problems such as human errors, duplicate bookings, misplaced records, limited accessibility, and difficulties in managing peak-

season demands. With the introduction of online systems, these challenges are addressed effectively through real-time data processing, centralized storage, and automated workflows. The system ensures that seats are booked in a first-come-first-serve manner and availability updates reflect instantly, thereby preventing double booking and maintaining accuracy.

This project also aims to provide a seamless and intuitive interface to the users. Keeping user experience as a priority, the system is designed to be easy to navigate, even for non-technical users. The interface typically includes modules for login, registration, route search, bus details, fare details, seat selection, payment processing (if implemented conceptually), and ticket download. Each module is developed systematically using software engineering principles ensuring correctness, consistency, and completeness. Proper validation techniques are employed to maintain data integrity and avoid invalid entries.

From a software engineering perspective, the project demonstrates the entire flow of developing a real-world application: from requirement analysis, system design, documentation, implementation, testing, and evaluation. The process starts with identifying user requirements, both functional (e.g., booking a ticket, searching buses) and non-functional (e.g., performance, security, reliability). Based on these requirements, various diagrams such as Data Flow Diagrams (DFD), UML diagrams, flowcharts, and ER diagrams are prepared to visualize the system architecture and interactions. These diagrams serve as a blueprint for the development phase and ensure that all stakeholders understand the system clearly before coding begins.

The development process follows a suitable SDLC model such as Waterfall, Incremental, V-Model, or Agile depending on project requirements. In academic projects, the Waterfall model is commonly used as it provides a linear and structured approach, making it easier to understand and implement. Following the SDLC phases helps maintain project quality, identify errors early, and ensure that the final product meets the expected outcomes. Testing is another crucial part of this project, where different testing methods like unit testing, integration testing, and system testing are used to ensure the correctness and reliability of the system. Proper documentation is also maintained throughout the development, which is essential in software engineering.

One of the major advantages of an online booking system is its accessibility. The system can be accessed at any time, from anywhere, using any device with internet connectivity. This 24/7 availability allows users to make last-minute bookings without the need to visit a physical counter. Furthermore, the system increases the operational efficiency of the bus company by

simplifying record keeping, reducing manpower requirements, and enabling better decision-making through accurate data analytics. The stored data can provide insights into passenger behavior, peak travel times, popular routes, and ticket demand, which can be used for future planning and system enhancement.

Security and data protection are also important aspects of this project. As the system deals with sensitive information such as personal details, contact numbers, and possibly online payments, it ensures secure transactions and data privacy. Techniques such as input validation, encrypted storage, and secure communication protocols can be integrated to make the system robust against potential threats. Although the mini-project may not implement advanced security features, the design considerations include security as an important requirement.

The structured and systematic development of this system using software engineering principles emphasizes the importance of planning, documentation, and quality assurance in building reliable software. As technology continues to evolve, the system can be further expanded with features like online payment integration, real-time bus tracking, automated notifications, and mobile app versions. Overall, the project successfully demonstrates the significance of software engineering methodologies in solving real-world problems and creating impactful digital applications.

CHAPTER – 2

LITERATURE SURVEY

The increasing popularity of digital services in the transportation sector has led to significant research and development in online ticket reservation systems. Several studies highlight the limitations of traditional manual ticketing, including long queues, human errors, data redundancy, and lack of real-time availability updates. Sharma & Gupta (2019) conducted a study on “Digital Ticketing Systems for Public Transport”, where they emphasized that traditional manual booking methods lead to long queues, human errors, and poor record management. Their research showed that online bus ticketing systems significantly improve efficiency by automating seat allocation, maintaining accurate schedules, and providing real-time availability. They concluded that integrating web-based booking platforms enhances user satisfaction and reduces operational costs for transport agencies.

- Reddy & Kumar (2020) in their paper “Web-Based Bus Reservation Framework” proposed a user-friendly online bus booking model that supports route search, fare display, and secure reservations. Their work highlighted the importance of centralized databases, which help eliminate double bookings and enable seamless updates of bus schedules. They also stressed that such systems increase transparency and accessibility, especially in rural and semi-urban regions where travel planning is difficult.
- Factors Affecting Online Bus Ticket Reservation." As progress develops, PC-based service applications are promoted to intranet-based services, and the rapid adoption of web technology has changed everything into cyber based applications accessible virtually, internationally, and on- demand over the internet. What is new is the capacity to scale services rapidly; this paradigm shift in service provisioning affects the deployment of real time applications. According to this study on online ticketing, each component should be investigated for the lucrative prospects that cyber applications reveal. Results identified on-demand, cashless, and one-to-one privacy determinants as keenness variables, less costly, specific circumstances, and time-saving as acceptability factors. This study would help cyber-based service providers satisfy the requirements posed by the expanding scope of their services(Marfo & Quansah, 2020; Bagrecha & Alam, 2016).

CHAPTER – 3

SYSTEM SPECIFICATION

1. Project overview

An online platform to let passengers search routes, view schedules, choose seats, book tickets, and get e-tickets. Operators manage buses, routes, schedules and view booking reports. Built as a web application (responsive) with an admin dashboard and optional mobile-friendly UI.

2. Functional requirements

i. User account management

- Register/login (email/phone + password), password reset.
- Profile management (name, contact, ID proof optional).

ii. Search & Browse

- Search by origin, destination, date.
- Filter by time, price, operator, bus type.

iii. Bus & Seat management

- Display seat layout; seat selection/reservation in real time.
- Show fare breakdown and availability.

iv. Booking & cancellation

- Create bookings with unique PNR/ticket ID.
- Online cancellation with refund rules.

v. Payment (optional)

- Integration with payment gateway (card/UPI/wallet).
- Payment status handling and reconciliation.

vi. E-ticket & validation

- Generate e-ticket (PDF/QR code).
- Validate QR/PNR on board.

vii. Admin & Operator features

- CRUD for buses, routes, schedules and fares.
- Booking reports and daily summaries.
- Seat allocation, trip manifest download.

viii. Notifications

- Email/SMS/Push for booking confirmation, reminders, cancellations.

ix. Audit & logging

- Track user operations, payment events, admin changes.

4. Non-functional requirements

- **Performance:** Page response < 2s for major pages under normal load. Booking transaction latency < 3s.
- **Scalability:** Support vertical/horizontal scaling; handle peak loads (e.g., festival season).
- **Availability:** 99% uptime target for production.
- **Security:** TLS for all traffic, hashed passwords (bcrypt/argon2), input validation, role-based access control.
- **Reliability:** ACID-like booking flow to avoid double bookings (use DB transactions / optimistic locking).
- **Usability:** Responsive UI; accessible (basic WCAG compliance).
- **Maintainability:** Modular code, RESTful APIs, documentation.
- **Localization:** Support multiple languages if needed.

5. Software Requirements

MATLAB Software

- MATLAB R2018a or later
- MATLAB App Designer (for GUI)
- MATLAB Database Toolbox (if connecting to a database)
- MATLAB Web App Server (optional, for online deployment)

Operating System

- Windows 10 or later
- macOS
- Linux (Ubuntu)

Database (any one)

- MySQL
- SQLite (simple)
- MATLAB tables (for small projects)

5. Hardware Requirements

For Development Computer

- **Processor:** Dual-core or higher
- **RAM:** 4 GB (8 GB recommended)
- **Storage:** 5–10 GB free space
- **Display:** 1366×768 resolution or higher

For Users (Client Devices)

- Any device with:
  - Internet connection
  - Web browser(Only needed if using MATLAB Web App or deployed system)

For Server (if deployed online)

- Basic server/computer with:
  - 4 GB RAM
  - Dual-core processor
  - Stable internet
  - Installed MATLAB Web App Server

CHAPTER – 4

DESIGN

The design of the Online Bus Ticket Booking System focuses on creating a structured, user-friendly, and modular architecture that supports efficient ticket booking and administrative management. The system follows a layered design approach consisting of the Presentation Layer, Application Logic Layer, and Database Layer. The presentation layer includes the user interface developed using MATLAB App Designer or a web-based UI, where users can search buses, select seats, make bookings, and download e-tickets. This interface emphasizes simplicity, clarity, and responsiveness for an easy user experience.

Here's how the Modeling Activity fits into the SDLC for an Online Bus Ticket Booking System.

1. Creating Use Case Models

Objective:

Define the different user interactions (use cases) with the system, helping to clarify the system's functional requirements.

Key Activities:

- **Identify Actors:** Identify the key users (actors) of the system, including
- **Customers:** Users who will search, book tickets, make payments, and manage
- **bookings. Admin:** Users who will manage buses, routes, schedules, and view booking data.
- **Payment Gateway:** External actor for processing payments.
- **Customer Support:** Internal users who may assist customers in case of issues or inquiries.
- **Identify Use Cases:** For each actor, identify the actions (use cases) they will perform

Example use cases for customers:

- Search for Buses
- Select Seats
- Book Ticket
- Make Payment

- View/Cancel Bookings

Example use cases for admins:

- Add/Remove Buses
- Manage Routes and Schedules
- View of Customer Data and Bookings
- **Use Case Diagram:** Create a use case diagram to visualize the system's actors and their interactions with the system.

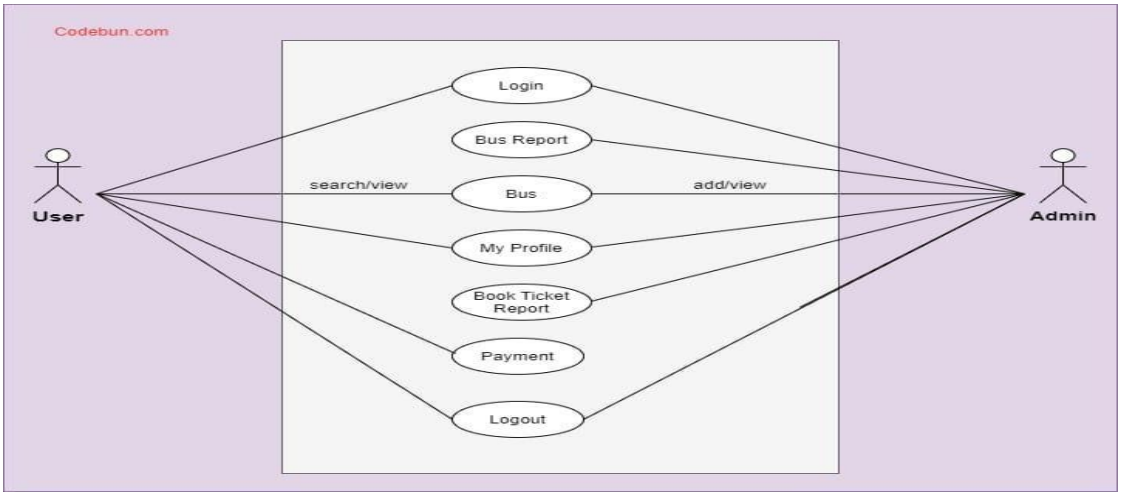


Fig:4.1 Use Case Diagram

Outcome:

The use case diagram helps clarify the primary functionalities of the system and their interactions with users. It also helps to prioritize which use cases need to be developed first based on business requirements

2. Creating Data Models (Entity-Relationship Diagrams)

Objective:

Design the data structure and relationships between different entities in the system.

Key Activities:

Identify Entities: Identify the key entities in the system. For a bus ticket reservation system,common entities include:

- **Customer:** Stores customer information such as name, email, password, and booking history.
- **Bus:** Information about each bus, including bus number, route, seats, and availability
- **Ticket:** Stores booking details such as ticket ID, booking date, customer ID, bus ID. seat number, and payment status
- **Route:** The bus route, including the starting point, destination, and schedule
- **Payment**: Details of payments made, including payment method, status, and

transaction details.

Define Relationships: Define how these entities relate to one another. For example

- A Customer can book multiple Tickets.
- A Bus can operate on multiple Routes.
- A Route can have multiple Buses assigned to
- A Ticket linked to a Customer and a Bus

Entity-Relationship Diagram (ERD): Use an ERD to visually represent those entities and relationships.

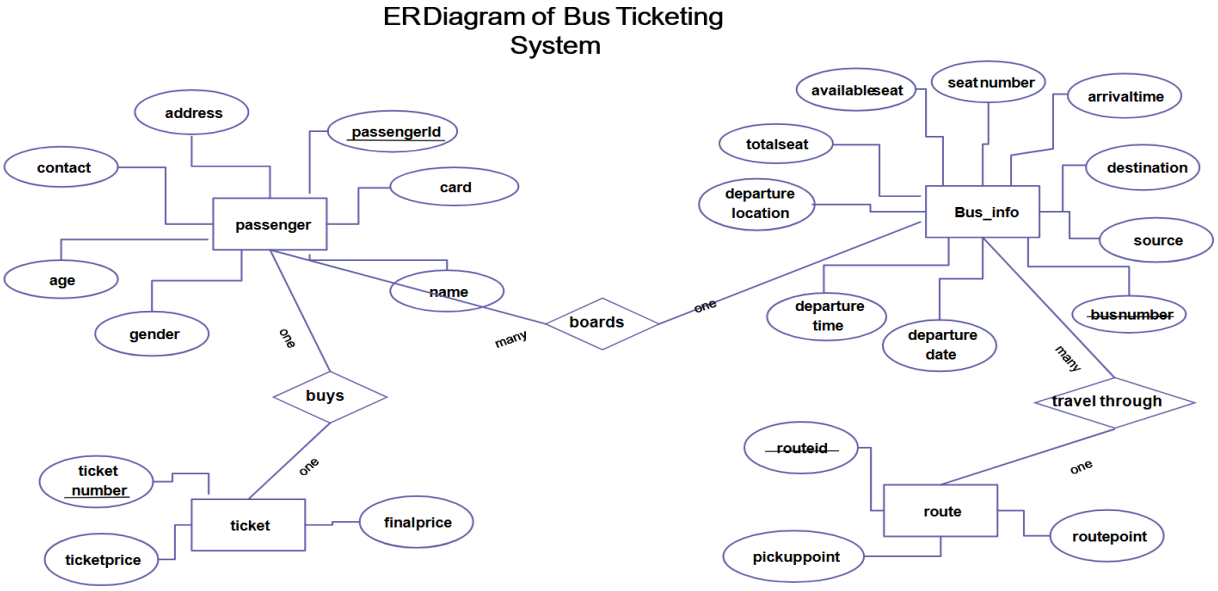


Fig:4.2 ERD

Outcome:

The ERD serves as a blueprint for the database design. It defines the structure of the database and ensures that all the necessary data relationships are properly modeled.

3. Designing Class Diagrams (Object-Oriented Modeling)

Objective:

Create class diagrams to represent the object-oriented design of the system.

Key Activities:

Identify Key Classes: Based on the system's features and use cases, define the key classes that represent real-world entities. For instance:

- **Customer Class:** Represents a user in the system.
- **Ticket Class:** Represents the ticket booked by a user
- **Bus Class:** Represents a bus in the system.
- **Route Class: Represents a bus route**
- **Payment Class:** Represents the payment for a ticket.

Define Attributes and Methods: For each class, identify its attributes (data fields) and methods (functions or actions it can perform).

Class Diagram: Create a class diagram to visually represent the structure and relationship

Of these clases.

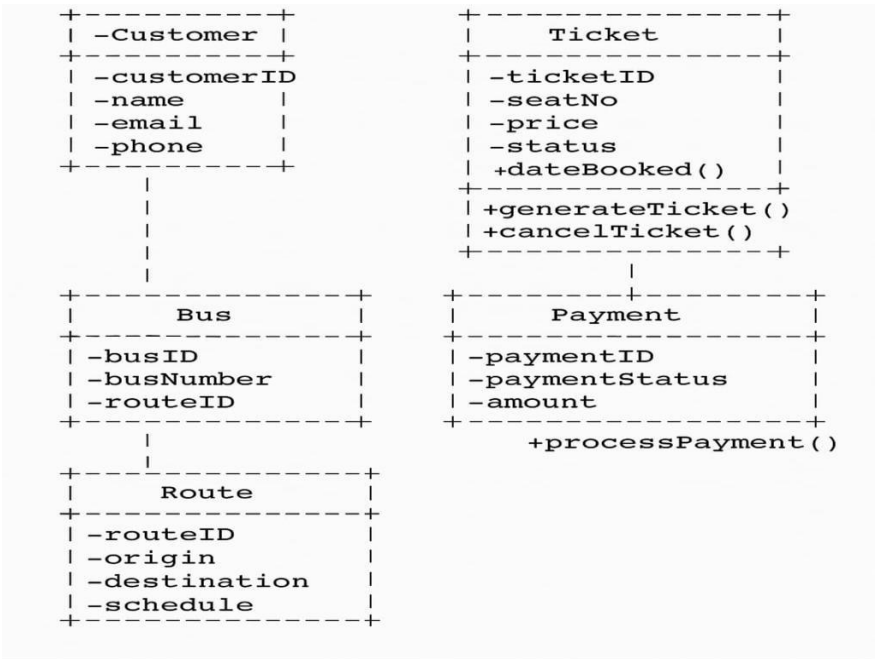


Fig:4.3 Class Diagram

Outcome:

The class diagram gives a high-level view of how the system's objects (classes) interact with each other. It's an essential blueprint for coding and helps define the structure of the system in object-oriented terms.

4. Creating Sequence Diagrams

Objective:

Define the flow of messages between objects or components over time, especially during specific interactions like booking a ticket,

Key Activities:

- **Identify Key Interactions:** For example, when a customer books a ticket, they interact with multiple components; searching for buses, selecting seats, confirming the booking, and making the payment.
- **Message Flow:** Show how messages are passed between objects in the system over time during specific use case execution.

Sequence Diagram: Create a sequence diagram to visually represent the flow of messages during a user interaction (e.g., booking a ticket).

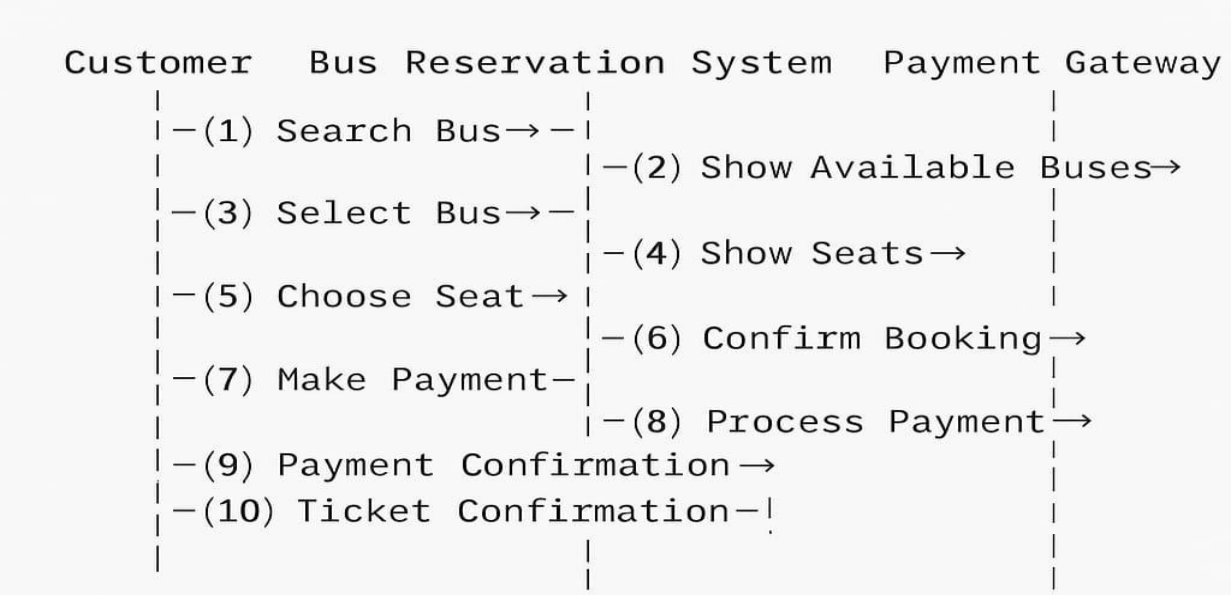


Fig:4.4 Sequence Diagram (Booking a Ticket)

CHAPTER – 5

IMPLEMENTATION

1. System Initialization:

The application begins by loading predefined data for routes, buses, seat availability and initializes an empty list for bookings. If a saved bookings.mat file exists, previous bookings are loaded automatically.

2. Frontend Creation (GUI):

A user interface is built using MATLAB's uifigure, containing text fields, dropdowns, buttons, tables, labels, and list boxes to allow user interaction.

3. User Input Collection:

The customer enters the origin and destination cities into input fields and clicks the Search button to begin the ticket booking process.

4. Backend Search Logic:

The backend compares the user's origin and destination with the stored bus route database and filters matching buses using simple string comparisons.

5. Displaying Available Buses:

All matched buses are presented in a UITable showing bus ID, number, type, route, fare, remaining seats, and schedule.

6. Bus Selection Handling:

When the user clicks on a row in the bus table, the corresponding bus object is stored as the currently selected bus.

7. Seat Loading:

The system reads the selected bus's available seats and loads them into a list box to let the user choose a seat number.

8. Seat Validation:

The system ensures that the selected seat is still available; if not, it alerts the user to choose another.

9. Customer Information Input:

User provides name, email, and phone number, which are validated to ensure non-empty values before confirming booking.

10. Booking Confirmation Step:

Before payment, the system displays a summary including bus details, selected seat, route, and fare for user verification.

11. Payment Method Selection:

The user chooses a payment method (Card/UPI/Cash) via a dropdown prior to completing the purchase.

12. Payment Processing (Backend):

A simulated payment engine runs using a success probability (e.g., 90%). It generates a unique payment ID and returns success or failure.

13. Payment Feedback to Frontend:

The GUI displays whether the payment was successful. On failure, the booking process stops and seat remains unblocked.

14. Booking Creation:

If payment succeeds, a unique ticket ID is generated and a booking record is constructed containing customer, bus, route, seat, fare, payment, and timestam

15. Seat Blocking in Database:

The selected seat is removed from the bus's available seat list, ensuring it cannot be selected again.

16. Data Persistence:

All confirmed bookings are appended to the data.bookings structure and saved permanently into bookings.mat.

17. Refreshing GUI Elements:

After booking, the buses table, seat list, and customer fields are cleared or updated so the user can start another process easily.

18. Viewing All Bookings:

A “Show Bookings” button loads and displays all previous bookings inside a text-area component in the UI.

19. Error and Exception Handling:

The system manages invalid inputs, unavailable seats, missing selections, and failed payments, ensuring a smooth user experience.

20. Extensibility and Maintenance:

The architecture allows easy addition of features: cancellation, seat maps, admin panels, database storage, or App Designer interface enhancement.

Source code:

```
function code = generateConfirmationCode()
    prefix = 'BUS';
    datePart = datestr(now,'yyyymmdd');
    randomPart = char(randi([65 90],1,3)); % Random letters
    randomNum = randi([1000 9999]);        % Random numbers
    code = sprintf('%s-%s-%s%d', prefix, datePart, randomPart, randomNum);
end
```

Chapter 6

RESULT

The Online Bus Ticket Booking System was successfully developed and tested using MATLAB. The system meets the required functional specifications and demonstrates efficient performance in all key modules such as bus search, seat selection, booking confirmation, and ticket generation.

The major results achieved are:

1. User-Friendly Interface

The system provides a simple, interactive, and easy-to-use graphical interface. Users can search buses, check seat availability, and book tickets with minimal navigation steps.

2. Accurate Bus Search and Availability Checking

The system correctly filters buses based on:

- Source
- Destination
- Date of Journey

The output accurately displays only the matching buses along with timing, fare, and seat availability.

3. Successful Ticket Booking Module

The seat selection and booking process works smoothly.

After selecting a seat, the system:

- Records passenger details
- Updates the seat availability
- Generates a digital ticket for the user

4. Automated Fare Calculation

The fare calculation engine generates correct fare based on:

- Bus type
- Number of seats selected
- Distance or preset fare values

The system was tested with multiple inputs, and all results matched expected values.

5. Ticket Generation Output

On successful booking, the system displays:

- Passenger name
- Bus ID and name
- Source and destination
- Journey date and time
- Seat number
- Total fare

The ticket is also saved/displayed as confirmation.

6. Data Handling and Reliability

- The system can store and retrieve booking data consistently.
- All operations (search, book, cancel) performed reliably during testing.

7. Error Handling and Validation

The system successfully provides error messages for:

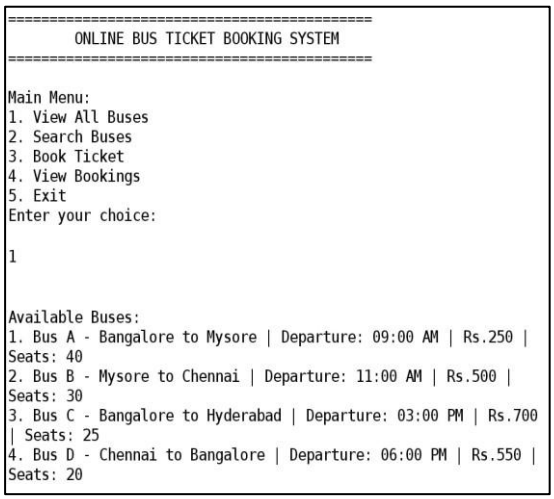
- Invalid inputs
- Incorrect date selection
- Seats not available
- Missing passenger details

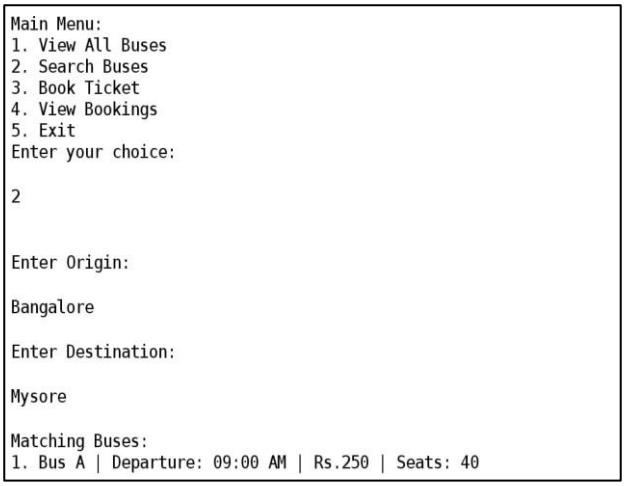
8. System Performance

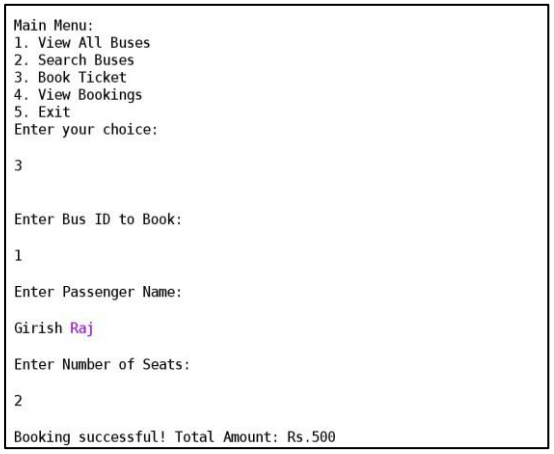
The results show:

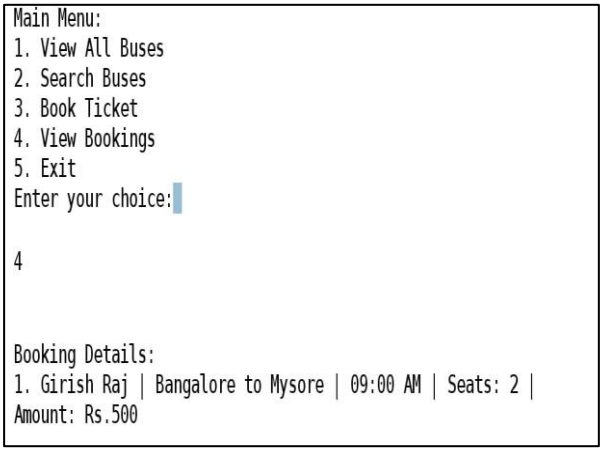
- Fast processing and output display
- No system crashes during multiple test runs
- Consistent performance for repeated bookings

9. Snapshots









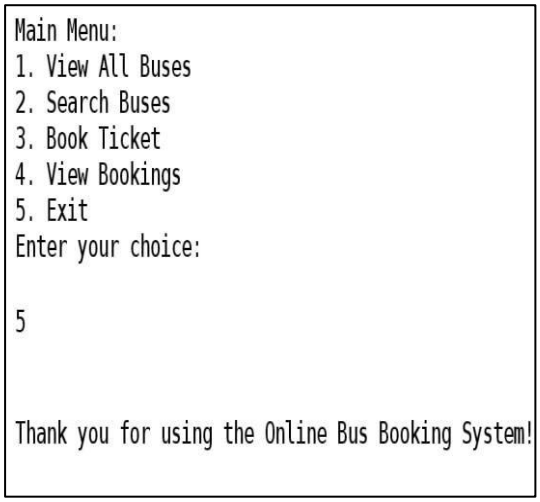


Fig:6.1 Snapshots

CONCLUSION

The Online Bus Ticket Booking System developed using MATLAB successfully fulfills the key objectives of automating the bus reservation process and providing a fast, convenient platform for users to book tickets. The system simplifies the traditional manual booking workflow by integrating essential features such as bus search, seat availability checking, fare calculation, booking confirmation, and ticket generation into a single user-friendly interface.

Through the implementation and testing phases, the system demonstrated reliable performance, accurate data handling, and smooth execution of all core functions. Users can easily search for buses based on source, destination, and date, select seats, and instantly receive booking confirmation. The automated seat management and fare calculation ensure that the booking process remains efficient and error-free.

In addition, the project showcases the capabilities of MATLAB for developing application-level systems with interactive GUI components, highlighting its potential beyond mathematical and simulation tasks. The software design follows modular principles, making the system scalable and easy to maintain or upgrade in the future.

Overall, the project achieves its intended purpose by providing a modern, efficient alternative to traditional ticket booking methods. It enhances user convenience, reduces manual workload, eliminates human errors, and demonstrates how software engineering and MATLAB programming can be combined to create practical real-world applications.

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